

# Sivani Bangaru

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## SUMMARY

**Data Engineer with 3+ years of experience** building scalable **data pipelines** and supporting enterprise **healthcare analytics** and **operational reporting** environments. Proficient in **Apache Spark, Apache Kafka, Apache Airflow, SQL, Python, AWS, and Azure** for **distributed data processing, streaming ingestion, cloud-based workflows, and data warehousing** solutions. Experienced in **data modeling, data validation, ETL development, and reporting support** using **Tableau and Power BI** across structured and unstructured data environments.

## PROFESSIONAL EXPERIENCE

### Data Engineer | Optum | IND | Aug 2023 - Aug 2024

- Built a distributed data pipeline using **Apache Spark** and **Delta Lake** that processed **400K+ clinical records** for **Scikit-learn** models, cutting processing latency by **10 minutes** across **2** analytics platforms.
- Configured real-time data streaming infrastructures by deploying **Apache Kafka** pipelines to ingest **10K+ patient embedding vectors** into an **OpenSearch** cluster through cross-functional team **peer-programming**, reducing lag by **15 minutes**.
- Tuned retrieval speeds for a **RAG GenAI** medical application by engineering text-chunking workflows using **LangChain** to process **200K+ unstructured medical documents**, saving **5 hours** of weekly downtime.
- Sped up data warehouse query runtimes by **3 hours** weekly by writing optimized **SQL queries** inside **Amazon Athena** over **300K lines** of unstructured healthcare logs stored in **Amazon S3** repositories.
- Fixed **8 hours** of weekly dashboard rendering delays by applying database **schema normalization** inside an on-premise **PostgreSQL** engine powering enterprise **Power BI** dashboards tracking **200K+ patient health updates**.
- Migrated **4** production environments to ensure strict compliance with **HIPAA regulations** by moving legacy **FHIR interoperability** databases containing **150K+ patient identifiers** to structured **Data Marts** via **Apache Airflow** DAGs.

### Data Analyst | Dell Technologies | IND | Jul 2021 - Aug 2023

- Analyzed supply chain and customer support datasets using **SQL** and **Tableau** to generate **10 weekly analytical reports** supporting inventory forecasting, service performance tracking, and business decision-making activities.
- Streamlined recurring data reconciliation and reporting processes using **Excel Power Query, VLOOKUP, and Pivot Tables** to reduce manual data preparation efforts by **10 hours weekly** across business analytics functions.
- Audited and cleansed transactional datasets using **MySQL, Python Pandas**, and structured data validation techniques to improve reporting accuracy across **8 business intelligence dashboards** used for operational analysis.
- Coordinated data validation and issue tracking activities through **JIRA, SQL Server Management Studio (SSMS)**, and reporting documentation to maintain data consistency across **5 analytics reporting workflows**.
- Developed **Tableau Prep, KPI scorecards**, and data visualization workflows with senior analysts to improve reporting consistency across **6 monthly business review cycles**.
- Monitored operational datasets using **Azure Data Factory** and **Azure SQL Database** to support automated data integration workflows across **7 enterprise reporting pipelines**.

## TECHNICAL SKILLS

- Programming & Query Languages:** SQL, Advanced SQL, Python, PySpark, Shell Scripting, JSON, YAML
- Data Engineering & ETL:** ETL/ELT Pipelines, Apache Airflow, Apache Spark, Databricks, Apache Kafka, Hadoop, Hive, Delta Lake, Data Integration, Data Transformation, Batch Processing, Data Modeling, Star Schema, Snowflake Schema
- Cloud Platforms:** AWS (S3, Glue, Lambda, Redshift, Athena, CloudWatch, IAM), Azure (Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure Blob Storage, Azure SQL Database), GCP (BigQuery, Google Cloud Storage)
- BI & Visualization Tools:** Tableau, Power BI, Looker, SSRS
- Healthcare & Compliance:** Healthcare Analytics, HIPAA Compliance, EHR/EMR Data Processing, Clinical Data Reporting, Claims Data Analysis, HL7, Healthcare Data Integration
- Databases:** Snowflake, SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, Teradata
- Modern Data Stack:** dbt (Data Build Tool), Fivetran, Alteryx, Data Lakes
- DevOps & Version Control:** Git, GitHub, Docker, Jenkins, CI/CD Pipelines, Kubernetes, JIRA
- AI/ML & Advanced Data Processing:** Machine Learning Data Preparation, Feature Engineering, Predictive Analytics, NLP Data Processing, Vector Databases, RAG Pipelines, LLM Data Workflows, AI Data Pipelines

## EDUCATION

Master of Science in Data Science - **Montclair State University, New Jersey, USA**

Aug 2024 - May 2026

## PROJECTS

### Business Performance Reporting & Analytics System

- Developed a **SQL-based** reporting platform using **Excel Power Query** and **Python**, consolidating **12+ operational datasets** into automated workflows with **anomaly detection** across monthly business performance reporting cycles.